



Datathon 2026: What's Cooking



Data For Dinner Team

Track: Eat
Food Insecurities

Your Chefs



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Menu Theme (Problem Statement)

The Women in Data (WiD) Datathon's main challenge is exploring the world's food systems through data, from farm to fork.



Data For Dinner Team

- Our focus is on **food insecurity**.
- Our scope: the globe, the United States, the Southeast, and **Alabama**.
- **Food insecurity** impacts families & **diminishes** quality of life.
- Families face an **uneven** menu of **limited access**, **unaffordability**, and **undernourishment**.
- We **need** community-driven solutions that deliver equitable food *and* nourishment **for all**.



Diners

Stakeholders

Stakeholders

The stakeholders are the **local community** and **policy makers**.

- The members of the local community
 - Families
 - Farmers / Home Growers
 - Farmers Markets
 - Local Fishers
- Support from policy makers
 - Town government
 - County government
 - State government
 - Federal government



A kitchen scene with a white ceramic pitcher, wooden spoons, a potted herb plant, and a plate of mushrooms on a light-colored countertop.

Chef's Tools

Methodology & Approach

Food Insecurity Data Sources

Food and Agriculture Organization of the United Nations

- Suite of Food Security Indicators
- <https://www.fao.org/faostat/en/#data/FS>
 - Regions: Africa, Asia, Europe, Latin America and the Caribbean, Oceania, United States of America, and World
 - Data Points: Severe food insecurity, Moderate food insecurity, Undernourishment, and Prevalence of unaffordability

Feeding America

- <https://map.feedingamerica.org/>
- Food Insecurity among the Overall Population in the United States (<https://map.feedingamerica.org/county/2024/overall/>)
- Food Insecurity among the Overall Population in Alabama (<https://map.feedingamerica.org/county/2024/overall/alabama>)

Preparing the Meal



- **Processing & Analysis**

- Batch downloads
- Merged datasets
- Excel for analysis
- Mapchart.net for geographic visualization details
- Data cleanup where data did not align with aggregations
- Tableau Public dashboard

- **Tools**

- GitHub
- Google Drive
- Google Meet
- MS Teams

Kitchen Mishaps

Obstacles Encountered

- Shifted story from food access to food insecurity
 - Data not easily available at the local level
- We wanted to include food allergy impact analysis, however data available was not *useful* and not detailed enough to inform our analysis

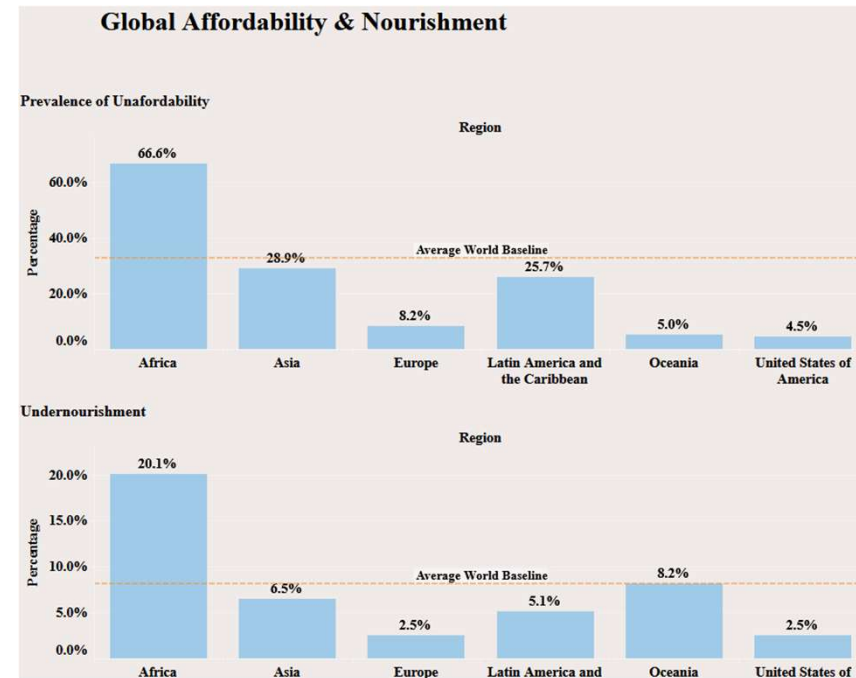
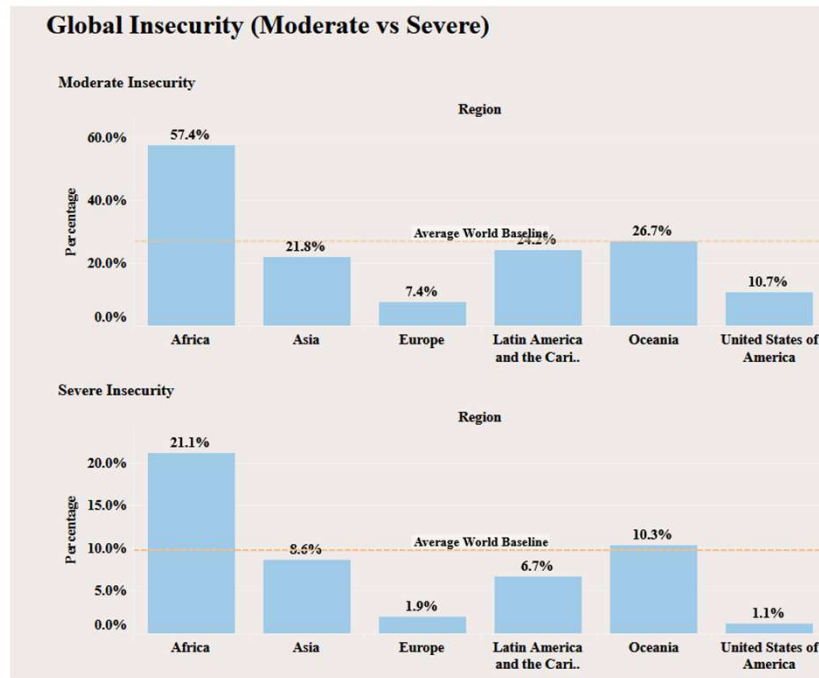




Appetizer

Global Food Insecurities

Global Food Insecurity



Globally, **millions experience** moderate and severe food insecurity due to poverty and various other factors.

Africa and Latin America have the largest populations of food insecurity.

Africa's food insecurity levels are moderate **57.4%**, and severe **21.1%**

Despite global baselines, the *distribution of food remains uneven*, leaving **entire populations** without food and nourishment.

Data from <https://www.fao.org/faostat/en/#data/FS>

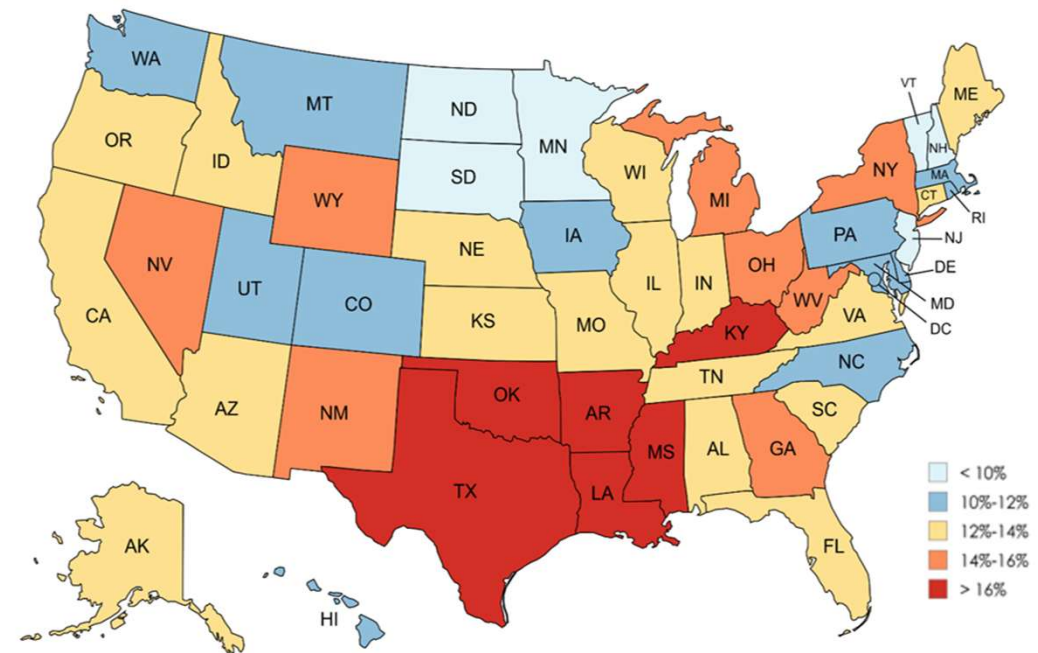
A top-down photograph of a rustic meal. A terracotta bowl filled with a red, chunky soup sits on a wooden board. A dollop of white cream is in the center, topped with fresh green herbs. Several pieces of dark, seeded bread are scattered around the bowl. A silver spoon lies on a dark, textured surface in the foreground. The entire scene is framed by a white, hand-drawn style border.

Soup

United States Food Insecurities

United States Food Insecurities

- **Food insecurity** is **defined** by the United States Department of Agriculture (USDA) as a household-level economic and social condition of limited or uncertain access to adequate food.
- Whereas food-secure households have consistent access throughout the year to adequate food for active, healthy living for all household members, **food-insecure** households lack that access at some time during the year.
- **Food insecurity** is often linked to one or *multiple factors*, creating a cycle that can be difficult to break.



Data from feedingamerica.org, map created with mapchart.net

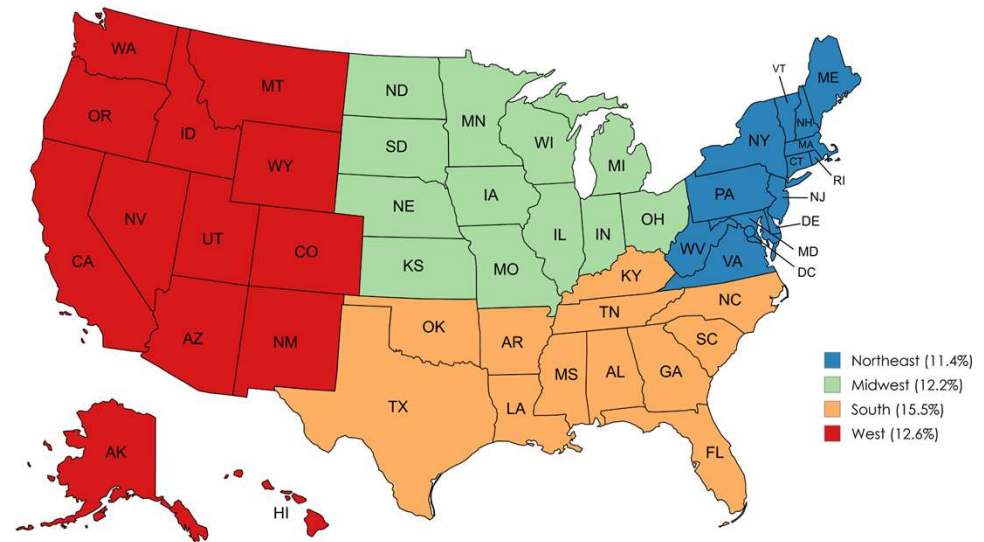
United States Food Insecurities by Region

Regional differences create pockets of heightened vulnerability.

Most notably in the **South** at **15.5%** compared to

- Northeast 11.4%
- Midwest 12.3%
- West 12.6%

Region	States
Northeast	CT, DE, DC, ME, MD, MA, NH, NJ, NY, PA, RI, VT, VA, WV
Midwest	IL, IN, IA, KS, MI, MN, MO, NE, ND, OH, SD, WI
South	AL, AR, FL, GA, KY, LA, MS, NC, OK, SC, TN, TX
West	AK, AZ, CA, CO, HI, ID, MT, NV, NM, OR, UT, WA, WY



Data from feedingamerica.org, map created with mapchart.net

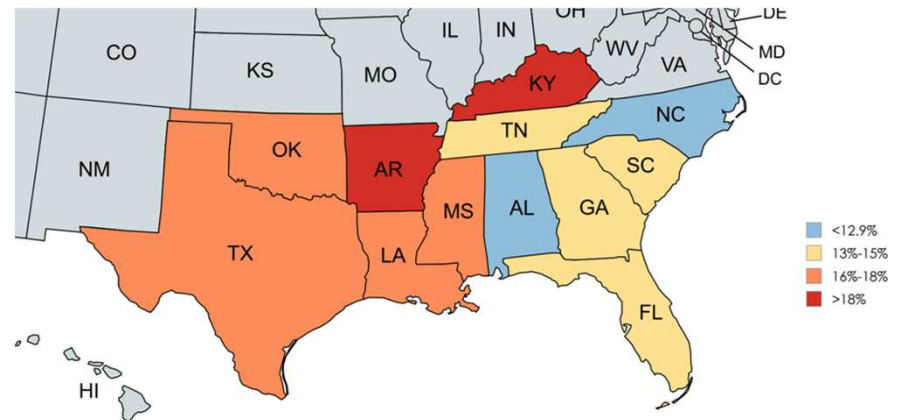
The background of the slide is a close-up photograph of various green leafy vegetables, likely lettuce or spinach, showing their veins and textures. The image is slightly blurred and has a dark green tint.

Salad

US Southeast Food Insecurities

US Southeast Food Insecurities

- States across the US **Southeast** have **higher** food insecurity **15.8%** than the US Average of 12.9%
- *Multiple factors* for families are:
 - High poverty
 - Rural geography
 - Weaker retailer networks
 - Transportation barriers
 - Demographic vulnerabilities
 - Climate variations (crops)
 - Policy gaps



The most affected states are Arkansas 19.4%, Kentucky 18.8%, and Louisiana, Mississippi, Texas, and Oklahoma with 17%.

A background image of a dining table with several plates of food, including salads and vegetables. Hands are visible using forks and knives. The image is overlaid with a white rectangular frame.

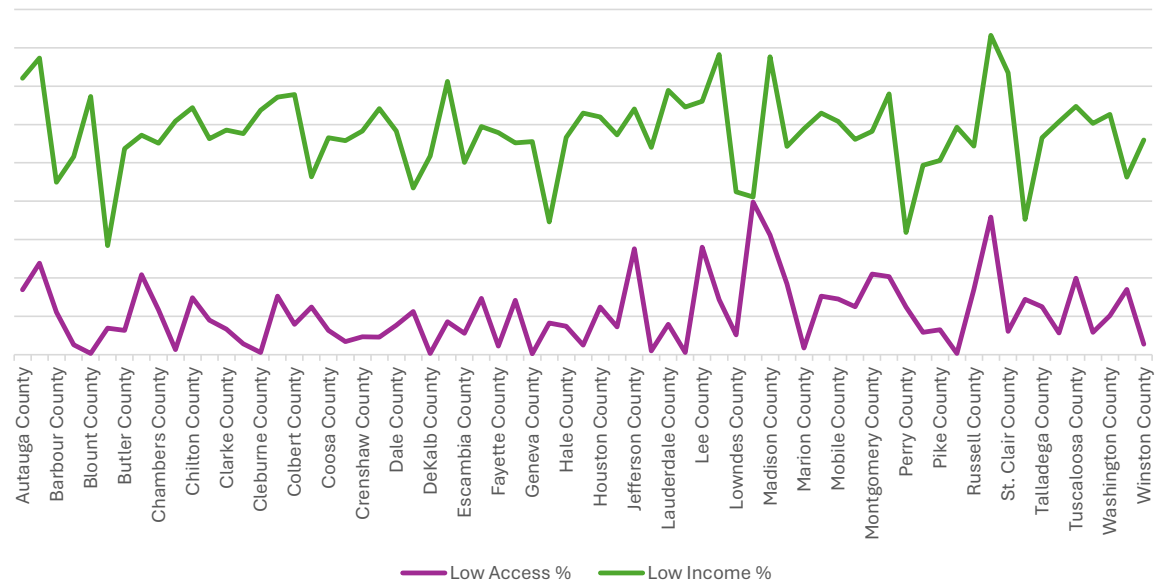
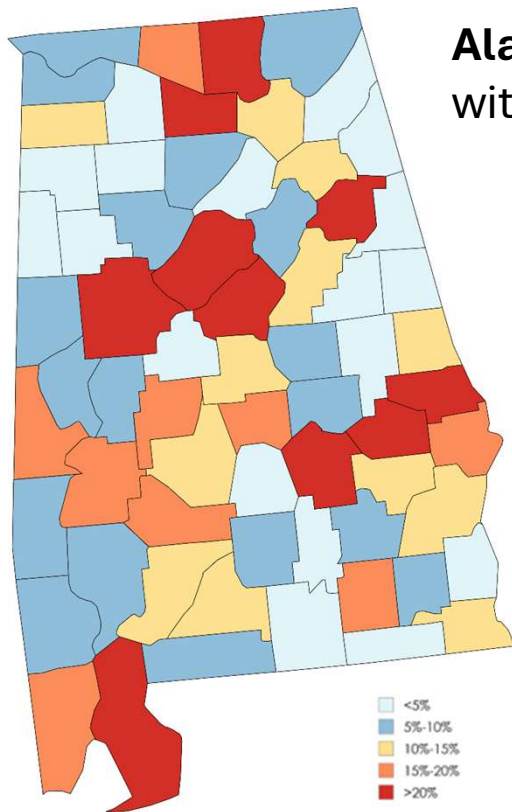
Main Course

Alabama Food Insecurities

Alabama Food Insecurities

Alabama faces *significant disparities* in **low access & low income**, with some counties experiencing extreme levels of **both**.

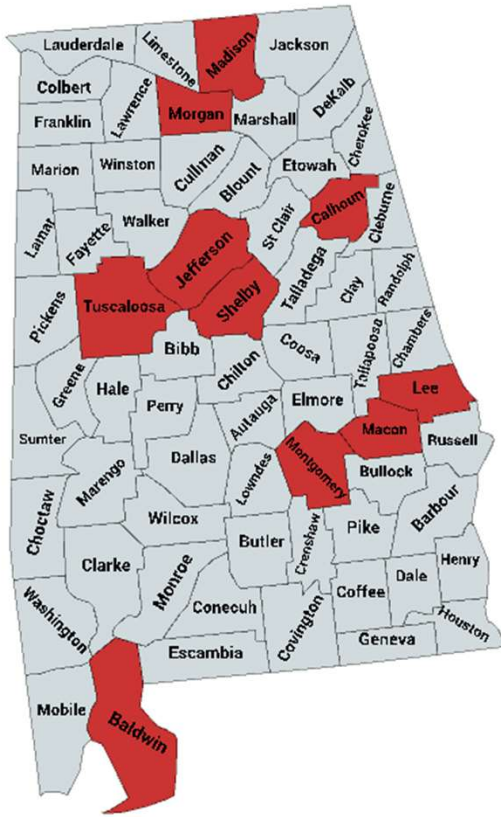
Low Access & Low Income Against Total Population



Data from feedingamerica.org, map created with mapchart.net

Alabama Regional: Highest Counties

Poverty areas, **rural** regions, **transportation** gaps, and **economic constraints** intensify the challenge, *creating a cycle* that is difficult to break.



Rank	County	Low-Income Low-Access	Key Drivers
1	Shelby	36.70%	Suburban sprawl, limited SNAP stores
2	Madison	35.30%	Rapid growth, uneven retailer spread
3	Lee	32.70%	Rapid population growth, uneven retailer access
4	Macon	31.10%	Deep-seated rural poverty, lack of grocery infrastructure, economic disinvestment
5	Jefferson	30.80%	Dense urban poverty, retailer clustering
6	Montgomery	25.40%	Urban-rural divide, high poverty pockets
7	Tuscaloosa	25.10%	Student-driven demand, low-income tracts
8	Baldwin	23.90%	Tourism economy, seasonal employment
9	Calhoun	21.90%	Industrial decline, limited fresh-food outlets
10	Morgan	21.70%	Rural geography, localized poverty, and economic barriers

Data from feedingamerica.org, map created with mapchart.net

The background of the slide features a close-up photograph of several decorative dumplings, likely Japanese dango, resting on a light-colored wooden surface. The dumplings are shaped into various forms: one is a white duck with an orange beak, another is a pink lotus flower with a yellow center, a third is a green and white dumpling with dark red berries, and others include a pink lotus and a white flower with a green garnish. The entire scene is framed by a thin white border.

Dessert

Potential Solutions

Local Community Solutions

- Communities can make rapid progress on **food insecurity** by focusing on practical, community-driven solutions that leverage existing infrastructure.
- These interventions are scalable, cost-effective, and aligned with **rural and urban** realities:
 - Support local food pantries
 - Support local shelter
 - Food pantry shelf stable staples (like a little library)
 - Local community small buses acting like a fresh pantry
 - Support for farmers and fishers

A rustic wooden table with a weathered surface. On the table, there is a light-colored ceramic coffee mug with a dark rim and handle, and a dark-colored ceramic pitcher with a handle. A wooden spoon lies horizontally across the table in front of the mug. The background is a plain, light-colored wall.

After Dinner Coffee

Future Work

Food Allergens impact on food insecurity

Food allergy impact analysis for food insecurity requires:

- continued research
- expanded data discovery
- deeper insights into: *
 - Affordability
 - Access
 - Nutrition

We identified significant food allergens specific to Alabama resources:

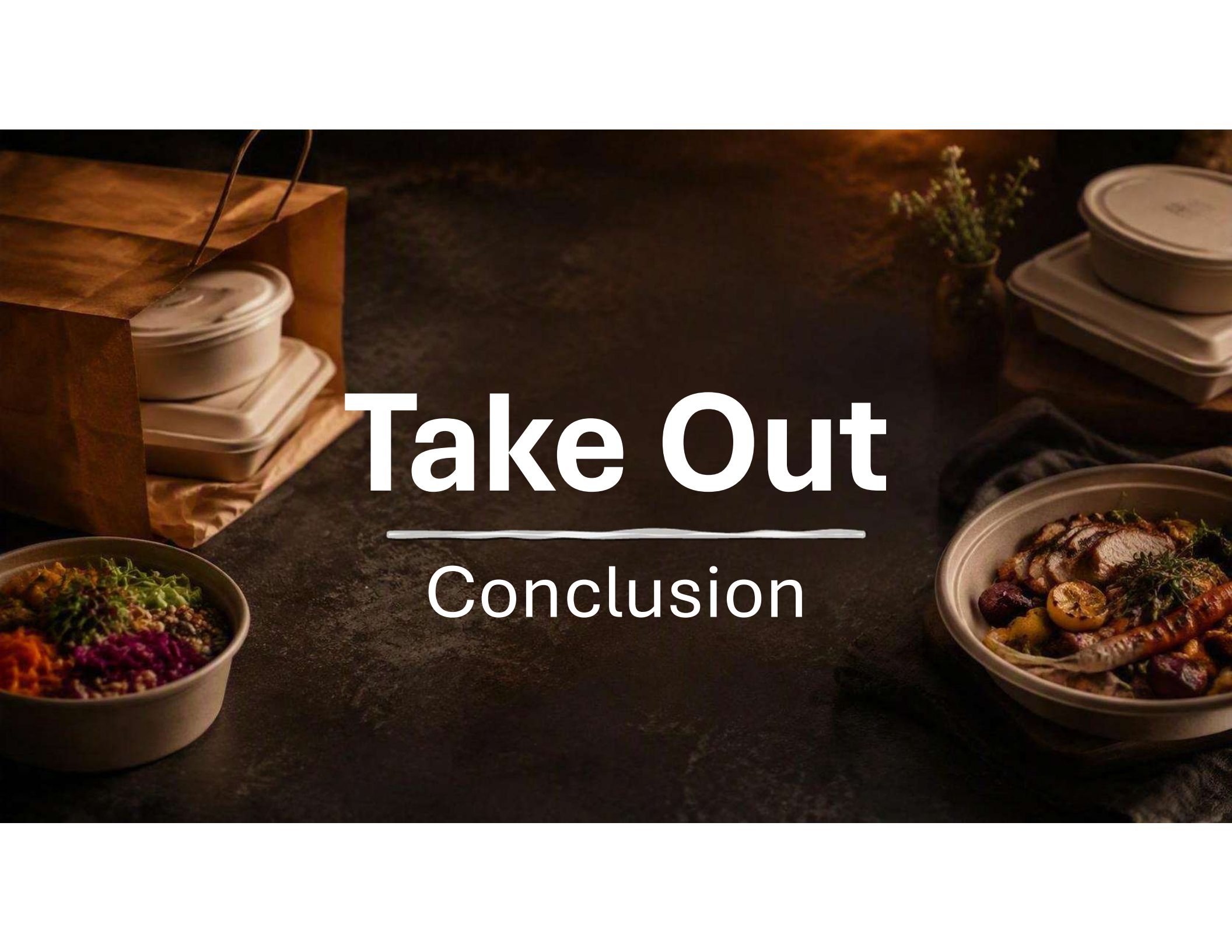
Seafood

- Shellfish allergy

Crops

- Peanuts (ranking 3rd in the nation)
- Soybeans
- Corn
- Pecans
- Wheat

*Especially considering food allergy constraints and local crop availability.

A dark, moody photograph of a table with food and takeout containers. In the foreground, there are two bowls of food: one with a colorful salad of shredded vegetables and another with a roasted dish featuring meat, carrots, and potatoes. In the background, there are stacks of white takeout containers and a brown paper bag. The lighting is warm and focused on the food, creating a cozy atmosphere.

Take Out

Conclusion

Take Aways

- Food insecurity is a multi-scale systems challenge that becomes clearer when analyzed through integrated, data-driven geospatial methods
- Structural instability, economic shocks, and limited food-system resilience create widespread vulnerability
- Rural isolation and limited food-retail infrastructure intensify insecurity
- Findings highlight the value of layered geospatial analysis for identifying actionable intervention points and support a practical framework of low-cost, scalable solutions



Come Back Again

GitHub Repo

Scan me!



Tableau Dashboard



Scan me!